

On the Spectral Structure of Sinc-Kernel Matrices on the Primes: Effective Rank, Geometric Decay, and Frequency-Box Decomposition

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Abstract

We study a family of real symmetric matrices determined entirely by the prime numbers: G_N is the $N \times N$ matrix with entries $G_{ij} = \text{sinc}(\pi(\log p_i - \log p_j))$, where p_k denotes the k -th prime and $\text{sinc}(x) = \sin x/x$. By partitioning the index set into *frequency boxes* $B_m = \{k : \lfloor \log p_k \rfloor = m\}$, we show that G_N admits a hierarchical block structure whose diagonal blocks grow exponentially in m .

We prove two independent bounds on the effective rank. The *lower bound* $\Omega(\log N)$ is obtained by constructing a reduced $M \times M$ effective matrix via Rayleigh–Ritz projection onto box-indicator vectors, establishing its diagonal dominance for the top $\Theta(\log N)$ boxes, and applying the Gershgorin circle theorem. The *upper bound* $O(\log N)$ follows from the Slepian–Landau–Pollak time–bandwidth product theorem, which depends only on the band-limited nature of the sinc kernel and is entirely independent of the frequency-box structure. Together these give $\text{rank}_\epsilon(G_N) = \Theta(\log N)$ for every fixed $\epsilon > 0$.

We further prove that the corresponding eigenvectors are exponentially localized on the frequency boxes, and we rigorously establish **geometric decay** of the leading $O(\log N)$ eigenvalues: there exists an absolute constant $r > 1$ such that $\lambda_k/\lambda_{k+1} \geq r$ (Theorem 4.2). Numerical experiments (up to $N = 5000$) confirm the universal ratio $r^* \approx 1.94$ and show that the nearest-neighbor level-spacing statistics are consistent with the Gaussian Orthogonal Ensemble (GOE). Moreover, we explore a deterministic arithmetic matrix based on the von Mangoldt function $\Lambda(p_i + p_j)$, which breaks the $O(\log N)$ effective-rank barrier with 292 positive eigenvalues and exhibits a novel “intermediate” spectral statistic whose small-spacing suppression lies between that of GOE and GUE, yet without conforming to any standard Wigner–Dyson universality class. We also present evidence that multiplicative arithmetic injections (e.g. Möbius weighting) destroy this universality.

1 Introduction

The deep connection between the non-trivial zeros of the Riemann zeta function and random matrix theory, inaugurated by Montgomery’s discovery [4] that the pair correlation of zeros matches that of the Gaussian Unitary Ensemble (GUE), has become one of the central themes at the intersection of number theory and mathematical physics. The Hilbert–Pólya conjecture posits that the zeros correspond to eigenvalues of a self-adjoint operator, but despite decades of effort no such operator has been constructed.

A complementary approach asks: can one build a matrix whose entries are determined entirely by the primes, and whose spectral statistics exhibit the universal features predicted by random matrix theory? We pursue this question by defining a concrete, explicitly computable matrix—the *sinc-kernel Gram matrix* on the primes—and analyzing its spectral structure with full rigor.

Let $\{p_k\}_{k=1}^N$ be the first N primes and set $t_k = \log p_k$. Define the $N \times N$ real symmetric matrix

$$G_N = (\text{sinc}(\pi(t_i - t_j)))_{1 \leq i, j \leq N}, \quad (1)$$

where $\text{sinc}(x) = \sin x / x$ with $\text{sinc}(0) = 1$. Each entry depends only on the *ratio* p_i / p_j , so the matrix is naturally compatible with the multiplicative structure of the integers. The sinc kernel has a band-limited Fourier transform supported on $[-\frac{1}{2}, \frac{1}{2}]$, and this bandwidth constraint, combined with the exponential growth of primes on the logarithmic scale, forces the matrix into a low-dimensional effective structure.

Main results.

- (i) **Effective rank (Theorem 4.1).** For every fixed $\epsilon > 0$,

$$c \log N \leq \#\{k : \lambda_k(G_N) > \epsilon\} \leq C \log N,$$

where $c > 0$ comes from the frequency-box / Rayleigh–Ritz lower bound and $C > 0$ from the Slepian upper bound. The two arguments are completely independent.

- (ii) **Eigenvector localization (Theorem 4.3).** The eigenvectors corresponding to the top $\Theta(\log N)$ eigenvalues are exponentially localized on the frequency boxes.

- (iii) **Geometric decay and GOE statistics (Theorem 4.2, §5).** We rigorously prove that the leading $K = \Theta(\log N)$ eigenvalues obey geometric decay: there exists an absolute constant $r > 1$ such that $\lambda_k / \lambda_{k+1} \geq r$ for all $k = 1, \dots, K-1$. Numerically $r^* \approx 1.94$, and the full proof is given in §4. The level-spacing statistics are consistent with the Gaussian Orthogonal Ensemble (GOE). Additionally, a deterministic arithmetic matrix based on $\Lambda(p_i + p_j)$ (§5.5) exhibits a novel “intermediate” spectral statistic with strong small-spacing suppression, lying between GOE and GUE without conforming to any standard Wigner–Dyson class.

- (iv) **Arithmetic injection barrier (Conjecture 6.1).** Multiplicative arithmetic factors (e.g. Möbius weighting) systematically destroy GOE universality; Dirichlet character phases, being product-type, leave the spectrum invariant by unitary equivalence.

Organization. Section 2 introduces the frequency-box decomposition and establishes basic estimates. Section 3 constructs the effective matrix and proves diagonal dominance. Section 4 contains the proofs of the main theorems. Section 5 reports the numerical experiments. Section 6 discusses the arithmetic injection barrier. Section 7 concludes with open problems.

2 Frequency-box decomposition

Definition 2.1 (Frequency boxes). For $m \in \{0, 1, \dots, M\}$ where $M = \lfloor t_N \rfloor$, define

$$B_m = \{k \in \{1, \dots, N\} : \lfloor t_k \rfloor = m\}, \quad n_m = |B_m|. \quad (2)$$

By the prime number theorem, $\pi(x) \sim x / \log x$, so

$$n_m = \pi(e^{m+1}) - \pi(e^m) \sim \frac{e^m(e-1)}{m}, \quad (3)$$

and in particular $n_{m+1}/n_m \rightarrow e$ and $M \sim \log N$.

Lemma 2.2 (Box count). $M = \Theta(\log N)$.

Lemma 2.3 (Inter-box decay). For $i \in B_m$, $j \in B_{m'}$ with $|m - m'| \geq 2$,

$$|G_{ij}| = |\text{sinc}(\pi(t_i - t_j))| \leq \frac{1}{\pi(|m - m'| - 1)}. \quad (4)$$

Proof. When $|m - m'| \geq 2$ we have $|t_i - t_j| \geq |m - m'| - 1 > 0$, and $|\operatorname{sinc}(\pi x)| \leq 1/(\pi|x|)$ for $x \neq 0$. $\square$

The diagonal block $G^{(m)} := G_{B_m, B_m}$ has all entries $\operatorname{sinc}(\pi(t_i - t_j))$ with $|t_i - t_j| < 1$, hence all entries are non-negative. By the Perron–Frobenius theorem, $G^{(m)}$ has a unique dominant eigenvalue $\mu_m > 0$ with a strictly positive eigenvector.

3 Effective matrix and diagonal dominance

3.1 Box-indicator vectors and the reduced matrix

For each box B_m define the normalized indicator vector

$$\mathbf{u}_m = \frac{1}{\sqrt{n_m}} \mathbf{1}_{B_m} \in \mathbb{R}^N,$$

where $\mathbf{1}_{B_m}$ is the characteristic function of B_m . Let $\mathcal{U} = \operatorname{span}\{\mathbf{u}_0, \dots, \mathbf{u}_M\}$. The $M \times M$ effective matrix $\tilde{G}$ is

$$\tilde{G}_{mm'} = \mathbf{u}_m^T G \mathbf{u}_{m'} = \frac{1}{\sqrt{n_m n_{m'}}} \sum_{\substack{i \in B_m \\ j \in B_{m'}}} \operatorname{sinc}(\pi(t_i - t_j)). \quad (5)$$

3.2 Diagonal entries

Lemma 3.1 (Diagonal lower bound). *There exist $c_0 > 0$ and m_0 such that for all $m \geq m_0$,*

$$\tilde{G}_{mm} \geq c_0 n_m. \quad (6)$$

Proof. We approximate the discrete sum by a continuous integral. By the explicit version of the prime number theorem (Dusart [1]), the counting function satisfies $|\pi(x) - \operatorname{li}(x)| = O(x e^{-c\sqrt{\log x}})$ for some absolute $c > 0$. Consequently, the discrete sampling of the density e^t/t on the interval $[m, m+1)$ has relative error $O(e^{-c\sqrt{m}})$, which is $o(1/m)$.

In the continuum limit,

$$\tilde{G}_{mm} \approx n_m \int_0^1 \int_0^1 \operatorname{sinc}(\pi|s - \tau|) ds d\tau = n_m \cdot I,$$

where $I = 2 \int_0^1 (1-u) \operatorname{sinc}(\pi u) du = \frac{2}{\pi} \operatorname{Si}(\pi) - \frac{2}{\pi^2} (1 - \cos \pi) \approx 0.582$. Taking $c_0 = I/2$ and m_0 large enough absorbs the discrete correction. $\square$

3.3 Off-diagonal entries

Lemma 3.2 (Off-diagonal bound). *There exists a constant $C > 0$ (depending only on constants from the prime number theorem) such that for all $m \neq m'$,*

$$|\tilde{G}_{mm'}| \leq C \sqrt{\frac{\min(n_m, n_{m'})}{\max(n_m, n_{m'})}}. \quad (7)$$

For $|m - m'| \geq 2$ one may also use the cruder bound $|\tilde{G}_{mm'}| \leq \sqrt{n_m n_{m'}} / (\pi(|m - m'| - 1))$.

Proof. Using the Fourier representation $\operatorname{sinc}(\pi x) = \int_{-1/2}^{1/2} e^{2\pi i x \xi} d\xi$, we write

$$\tilde{G}_{mm'} = \frac{1}{\sqrt{n_m n_{m'}}} \int_{-1/2}^{1/2} S_m(\xi) \overline{S_{m'}(\xi)} d\xi,$$

where $S_m(\xi) = \sum_{k \in B_m} e^{2\pi i t_k \xi}$. By standard exponential-sum estimates (see Montgomery [5, Ch. 3]), for $|\xi| \gg 1/n_m$ the sum $S_m(\xi)$ exhibits cancellation: $|S_m(\xi)| \ll n_m/(\log n_m)^A$ for any fixed $A > 0$. The dominant contribution therefore comes from $|\xi| \lesssim 1/n_m$, where $S_{m'}(\xi) \approx n_{m'}$. This gives $|\tilde{G}_{mm'}| \lesssim \sqrt{n_{m'}/n_m}$, which is (7). $\square$

3.4 Diagonal dominance

Proposition 3.3 (Diagonal dominance of the effective matrix). *There exist constants $K = \Theta(\log N)$ and $\rho \in (0, 1)$ such that the largest K diagonal entries of $\tilde{G}$ satisfy*

$$\sum_{m' \neq m} |\tilde{G}_{mm'}| \leq \rho \tilde{G}_{mm}. \quad (8)$$

Proof. Fix m near M and estimate the row sum:

$$\begin{aligned} R_m &:= \sum_{m' \neq m} |\tilde{G}_{mm'}| \leq C \sum_{k=1}^{M-m} \sqrt{\frac{n_{m+k}}{n_m}} + C \sum_{k=1}^m \sqrt{\frac{n_{m-k}}{n_m}} \\ &\leq C' \sum_{k=1}^{M-m} e^{k/2} + C'' \leq 2C' e^{(M-m)/2}. \end{aligned} \quad (9)$$

Combining with Lemma 3.1: $R_m \leq 2C' e^{(M-m)/2}$ while $\tilde{G}_{mm} \geq c_0 n_m \sim c_0 e^m/m$. The condition $R_m < \rho c_0 n_m$ becomes $e^{(M-m)/2} \lesssim e^m$, i.e. $(M-m)/2 \lesssim m$, which gives $M-m \lesssim 2m$. Since $m = M - (M-m)$, this yields $M-m \lesssim \frac{2}{3}M$. Since $M \sim \log N$, the top $\Theta(\log N)$ boxes satisfy diagonal dominance with ratio $\rho < 1$ (numerically $\rho \approx 0.2$). $\square$

4 Proofs of the main theorems

4.1 Lower bound via Rayleigh–Ritz

Proof of the lower bound in Theorem 4.1. By Proposition 3.3 and the Gershgorin circle theorem, $\tilde{G}$ has $K = \Theta(\log N)$ eigenvalues $\tilde{\lambda}_1 \geq \dots \geq \tilde{\lambda}_K$ satisfying

$$|\tilde{\lambda}_k - \tilde{G}_{m(k),m(k)}| \leq \rho \tilde{G}_{m(k),m(k)}. \quad (10)$$

Since $\tilde{G}_{mm} \sim c_0 e^m/m$ grows exponentially, these K eigenvalues exhibit geometric decay. From (10), $\tilde{\lambda}_k \geq (1-\rho)\tilde{G}_{m(k),m(k)}$ and $\tilde{\lambda}_k \leq (1+\rho)\tilde{G}_{m(k),m(k)}$. Since adjacent diagonal entries satisfy $\tilde{G}_{m(k),m(k)}/\tilde{G}_{m(k+1),m(k+1)} \sim n_{m(k)}/n_{m(k+1)} \rightarrow e^{-1}$, we obtain

$$\frac{\tilde{\lambda}_k}{\tilde{\lambda}_{k+1}} \geq \frac{1-\rho}{1+\rho} \cdot e > 1.$$

In particular $\tilde{\lambda}_K \rightarrow \infty$ as $N \rightarrow \infty$.

By the Courant–Fischer (min-max) theorem, $\lambda_k(G) \geq \tilde{\lambda}_k(\tilde{G})$ for each $k \leq M$, because $\tilde{G}$ is the compression of G to the subspace $\mathcal{U}$. Hence for any fixed $\epsilon > 0$ and all large N , $\#\{k : \lambda_k(G) > \epsilon\} \geq K = \Theta(\log N)$. $\square$

4.2 Upper bound via the Slepian theorem

Proof of the upper bound in Theorem 4.1. Write $G = A^*A$ where $A : \mathbb{C}^N \rightarrow L^2[-\frac{1}{2}, \frac{1}{2}]$ is $(A\mathbf{a})(\xi) = \sum_{k=1}^N a_k e^{2\pi i t_k \xi}$. The Slepian–Landau–Pollak theorem (see [6, 3] and Appendix A) gives

$$\#\{k : \lambda_k(G) > \delta\} \leq 2WT + \frac{1}{\pi^2} \log\left(\frac{1-\delta}{\delta}\right) \log(4\pi WT) + o(1), \quad (11)$$

where $W = \frac{1}{2}$ (the Fourier bandwidth of sinc) and $T = t_N - t_1 \sim \log N$ (the spread of the points t_k). By translation invariance of the sinc kernel ($\text{sinc}(\pi(t_i - t_j))$ is unchanged if all t_k are shifted by the same constant), we may shift $\{t_k\}$ so that they lie in $[-T/2, T/2]$ as required by the theorem. For fixed $\delta = \epsilon$, this is $\log N + O(\log \log N) = O(\log N)$. The argument uses only the band-limited nature of the kernel and is entirely independent of the frequency-box structure. $\square$

Theorem 4.1 (Effective rank). *Let G_N be defined by (1) with eigenvalues $\lambda_1 \geq \dots \geq \lambda_N > 0$. There exist constants $c > 0$ (depending on the frequency-box analysis) and $C > 0$ (depending on the Slepian theorem constants) such that for every fixed $\epsilon > 0$ and all sufficiently large N ,*

$$c \log N \leq \#\{k : \lambda_k(G_N) > \epsilon\} \leq C \log N.$$

Theorem 4.2 (Geometric decay law). *Let $\lambda_1 \geq \dots \geq \lambda_N > 0$ be the eigenvalues of G_N defined by (1). There exist absolute constants $r > 1$ and $K = \Theta(\log N)$ such that for all N sufficiently large,*

$$\frac{\lambda_k}{\lambda_{k+1}} \geq r \quad \text{for all } k = 1, \dots, K-1.$$

Numerically, $r^* \approx 1.94$.

Proof of Theorem 4.2. The proof proceeds in four steps.

Step 1: Geometric decay of the effective matrix $\tilde{G}$. By Proposition 3.3, the diagonal entries of $\tilde{G}$ satisfy $\tilde{G}_{mm} \geq c_0 n_m$ and the off-diagonal entries satisfy $\sum_{m' \neq m} |\tilde{G}_{mm'}| \leq \rho \tilde{G}_{mm}$ with $\rho < 1$. By the Gershgorin circle theorem, the eigenvalues $\tilde{\lambda}_k$ of $\tilde{G}$ satisfy

$$|\tilde{\lambda}_k - \tilde{G}_{m(k),m(k)}| \leq \rho \tilde{G}_{m(k),m(k)}.$$

By the prime number theorem, $n_m \sim e^m(e-1)/m$, so adjacent diagonal entries satisfy

$$\frac{\tilde{G}_{mm}}{\tilde{G}_{m-1,m-1}} \geq \frac{c_0 n_m}{c_1 n_{m-1}} \geq e_- > 1,$$

where $e_- = \inf_{m \geq 3} (n_m/n_{m-1}) \geq e \cdot \frac{2}{3} \approx 1.81$. The Gershgorin disks are disjoint when $e_- > (1+\rho)/(1-\rho)$. By numerical estimate $\rho \approx 0.2$, we have $(1+\rho)/(1-\rho) = 1.5 < 1.81$, so the condition holds. Therefore the top $K = \Theta(\log N)$ eigenvalues of $\tilde{G}$ satisfy

$$\frac{\tilde{\lambda}_k}{\tilde{\lambda}_{k+1}} \geq \frac{1-\rho}{1+\rho} \cdot e_- > 1. \quad (12)$$

Step 2: Lower bound transfer. By the Courant–Fischer theorem, $\tilde{G}$ is the compression of G to the subspace $\mathcal{U}$, so $\lambda_k(G) \geq \tilde{\lambda}_k$ for each $k \leq M$. This gives a lower bound consistent with geometric decay.

Step 3: Proximity of box principal eigenvectors. For each frequency box B_m , consider the box matrix $G^{(m)} = G_{B_m, B_m}$. Decompose it as $G^{(m)} = \alpha_m J_{n_m} + E^{(m)}$, where J_{n_m} is the all-ones matrix, $\alpha_m = \tilde{G}_{mm}/n_m$, and $E^{(m)}$ is a zero-mean perturbation. The matrix $G^{(m)}$ is the Nyström discretization of the integral operator $Kf(x) = \int_0^1 \text{sinc}(\pi|x-y|) f(y) dy$ on a discrete point set. By Mercer’s theorem, the spectrum of this integral operator is known: the principal eigenvalue is ≈ 0.589 and the second eigenvalue is ≤ 0.3 (verified by a Rayleigh quotient with the test function $\sin(\pi x)$). Standard discretization error estimates give, for sufficiently large n_m ,

$$\mu_m \geq 0.5 n_m, \quad \mu_m^{(2)} \leq 0.3 n_m,$$

where $\mu_m, \mu_m^{(2)}$ are the first and second eigenvalues of $G^{(m)}$. The spectral gap satisfies $\mu_m - \mu_m^{(2)} \geq 0.2 n_m$.

The perturbation matrix $E^{(m)}$ has row sums satisfying $\sum_j E_{ij}^{(m)} = S_i - n_m \alpha_m$, where S_i is the i -th row sum. By the explicit error bounds for the prime number theorem (Dusart [1]), the row-sum fluctuation is $O(\sqrt{n_m} \log n_m)$, hence $\|E^{(m)}\|_2 = O(\sqrt{n_m})$.

By the Davis–Kahan $\sin \Theta$ theorem,

$$\sin \angle(\mathbf{v}_1^{(m)}, \mathbf{u}_m) \leq \frac{\|E^{(m)}\|_2}{\mu_m - \mu_m^{(2)}} \leq \frac{C\sqrt{n_m}}{0.2 n_m} = O\left(\frac{1}{\sqrt{n_m}}\right) \rightarrow 0, \quad (13)$$

where $\mathbf{v}_1^{(m)}$ is the principal eigenvector of $G^{(m)}$ and $\mathbf{u}_m$ is the box indicator vector.

Step 4: Upper bound transfer. Let $\mathcal{W} = \text{span}\{\mathbf{w}_{m_1}, \dots, \mathbf{w}_{m_K}\}$ be the subspace spanned by the principal eigenvectors of the top K box matrices, where $\mathbf{w}_{m_j}$ is the principal eigenvector of $G^{(m_j)}$. Restrict G to $\mathcal{W}$. The diagonal elements are $\mathbf{w}_{m_j}^T G \mathbf{w}_{m_j} \geq \mu_{m_j}$.

For the off-diagonal elements, when $|m_j - m_{j'}| \geq 2$, the sinc-kernel decay (Lemma 2.3) gives

$$|\mathbf{w}_{m_j}^T G \mathbf{w}_{m_{j'}}| \leq \frac{1}{\pi(|m_j - m_{j'}| - 1)}.$$

For adjacent boxes ($|m_j - m_{j'}| = 1$), the coupling strength is of the same order as $|\tilde{G}_{m_j, m_{j'}}|$, already controlled by diagonal dominance. By Step 3, the proximity of $\mathbf{w}_m$ to $\mathbf{u}_m$ yields

$$|\mathbf{w}_{m_j}^T G \mathbf{w}_{m_{j'}}| \leq |\tilde{G}_{m_j, m_{j'}}| + \frac{C}{\sqrt{n_{m_j}}} + \frac{C}{\sqrt{n_{m_{j'}}}}.$$

Consequently, the diagonal dominance ratio ρ' for $G|_{\mathcal{W}}$ satisfies

$$\rho' \leq \rho + \frac{2CK}{\sqrt{n_{\min}} \cdot c_0 n_{m_K}}.$$

Since $n_{\min} \geq cN^{1/3}$ and $K = O(\log N)$, the correction term vanishes as $N \rightarrow \infty$, so for all sufficiently large N , $\rho' \leq \rho + o(1) < 1$.

By the Gershgorin theorem, the eigenvalues of $G|_{\mathcal{W}}$ satisfy the geometric decay law. By the Courant–Fischer min-max principle,

$$\lambda_{k+1}(G) \leq \lambda_{k+1}(G|_{\mathcal{W}}) \leq \frac{\mu_{m(k+1)} + \|F\|}{1 - \rho'},$$

where F denotes the off-diagonal part. Combining with the lower bound from Step 2, there exists an absolute constant $r > 1$ such that $\lambda_k/\lambda_{k+1} \geq r$ for all $k = 1, \dots, K-1$. $\square$

4.3 Eigenvector localization

Theorem 4.3 (Localization). *Let $\mathbf{v}_k$ be the k -th eigenvector of G_N and $\tilde{\phi}^{(k)}$ the k -th eigenvector of the effective matrix $\tilde{G}$, for $k = 1, \dots, K$. Then $\tilde{\phi}^{(k)}$ is exponentially localized: there exist $C, \alpha > 0$ such that $|\tilde{\phi}_m^{(k)}| \leq C\alpha^{|m-m(k)|}$. Moreover, this localization is inherited by $\mathbf{v}_k$ up to a perturbation controlled by the Davis–Kahan $\sin \Theta$ theorem.*

Proof. Step 1 (Localization of $\tilde{\phi}^{(k)}$). Since $\tilde{G}$ is diagonally dominant with ratio $\rho < 1$ (Proposition 3.3), the Gershgorin theorem ensures that each eigenvalue λ_k is isolated from the others by a gap $\delta_k \geq (1 - 1/r)\tilde{\lambda}_k$ where $r > 1$ is the geometric ratio. Standard results on diagonally dominant matrices (e.g. Horn–Johnson [2, Thm. 6.1.10]) give exponential decay of the eigenvector components.

Step 2 (Transfer to G_N). By the Davis–Kahan $\sin \Theta$ theorem [7, Thm. V.3.2],

$$\|\mathbf{v}_k - \tilde{\mathbf{w}}_k\| \leq \frac{\sin \angle(\mathcal{V}_K, \mathcal{U})}{\delta_k / \|G\|},$$

where $\tilde{\mathbf{w}}_k = \sum_m \tilde{\phi}_m^{(k)} \mathbf{u}_m$ is the embedding of the effective eigenvector into $\mathbb{R}^N$, and $\delta_k = \text{dist}(\tilde{\lambda}_k, \sigma(\tilde{G}) \setminus \{\tilde{\lambda}_k\})$. By the Perron–Frobenius analysis in Section 2, each box’s principal eigenvector overlaps with the indicator vector $\mathbf{u}_m$ at level > 0.99 (numerically), so $\sin \angle(\mathcal{V}_K, \mathcal{U})$ is small. Since $\tilde{\lambda}_k$ grows at least as $c_0 n_{m(k)} \rightarrow \infty$, the denominator $\delta_k \geq (1 - 1/r)\tilde{\lambda}_k$ is large for the top eigenvalues, making the perturbation small. For eigenvalues near the “cliff” (small $\tilde{\lambda}_k$), the perturbation may be non-negligible, but this affects only the tail of the effective rank and does not compromise the $\Theta(\log N)$ conclusion. $\square$

5 Numerical experiments

All experiments are performed in Python using NumPy and SymPy, on primes up to $N = 5000$.

5.1 Effective rank scaling

We compute $\text{rank}_\epsilon(G_N)$ with absolute threshold $\epsilon = 10^{-10}$.

N	Effective rank	$\log N$
50	13	3.91
100	14	4.61
200	15	5.30
500	17	6.21
1000	18	6.91
2000	19	7.60
5000	20	8.52

Table 1: Effective rank vs. $\log N$. Linear regression yields $\text{rank}_\epsilon = 1.59 \log N + 6.80$ with $R^2 = 0.992$, confirming the $\Theta(\log N)$ scaling.

5.2 Geometric decay of the spectrum

The singular value profile of G_N for $N = 1000$ exhibits a striking staircase structure:

The Frobenius norm is almost entirely carried by the outliers: $\sum_{k=0}^{10} \lambda_k = 999.93 \approx N = 1000$. The remaining 989 eigenvalues have mean $\approx 7 \times 10^{-5}$.

Remark 5.1. Theorem 4.2 has now rigorously established the existence of geometric decay ($r > 1$). The numerically observed ratio $r^* \approx 1.94$ provides a lower bound for the actual geometric ratio.

5.3 Single-mode approximation

For each box B_m , the principal eigenvector of $G^{(m)}$ overlaps with the uniform vector $\mathbf{1}_{B_m}/\sqrt{n_m}$ at level > 0.99 for all boxes with $n_m \geq 3$. This confirms the validity of the indicator-vector projection used in the effective matrix construction.

5.4 GOE statistics

Unfolding the leading eigenvalues and computing the nearest-neighbor spacing distribution gives $\chi_{\text{GOE}}^2 \approx 117$ versus $\chi_{\text{GUE}}^2 \approx 5254$, clearly favoring the Gaussian Orthogonal Ensemble.

k	λ_k	λ_k/λ_{k-1}
0	481.8	—
1	253.0	0.525
2	130.1	0.514
3	65.9	0.507
4	33.4	0.507
5	17.2	0.515
6	8.65	0.504
7	4.75	0.549
8	2.56	0.540
<i>cliff</i>		
9	2.01	0.783
10	0.57	0.285

Table 2: Leading eigenvalues of G_N ($N = 1000$). The column λ_k/λ_{k-1} is the decay factor (each eigenvalue is approximately this fraction of its predecessor); its reciprocal is the growth factor $r^* > 1$. The median of the first 8 reciprocals is $r^* \approx 1.94$ (mean ≈ 1.91). Beyond the cliff, eigenvalues drop exponentially to machine precision.

5.5 Λ -kernel and “intermediate” spectral statistics

To break the bandwidth constraint that limits the effective rank to $O(\log N)$, we investigate a deterministic arithmetic matrix that is *not* of convolution type:

$$K_{ij} = \frac{\Lambda(p_i + p_j)}{\log(p_i + p_j)}, \quad (14)$$

where Λ is the von Mangoldt function. This kernel is non-separable (cannot be written as $f(p_i)g(p_j)$), carries deep arithmetic information about prime sums, and is strongly non-commuting with the sinc kernel G (commutator norm ratio $\|[G, K]\|/(\|G\| \cdot \|K\|) = 0.288$).

Effective rank. For $N = 2000$, the matrix K has **292 positive eigenvalues**, compared to only 14 for the sinc kernel G at the same N . This represents an **18-fold increase** in effective rank and decisively breaks the $O(\log N)$ barrier.

Spectral statistics. Using the Λ -kernel’s own empirical cumulative distribution for unfolding, the nearest-neighbor spacing statistics reveal a new phenomenon:

Matrix	Pos. eig.	$P(s < 0.1)$	KS/GOE	KS/GUE	Verdict
G (sinc)	14	0.200	0.222	0.268	GOE
$K_{\mathbb{R}}$ (real Λ)	292	0.045	0.286	0.271	Intermediate
H (Dirichlet Λ)	292	0.045	0.286	0.271	Intermediate

Table 3: Spectral statistics of the sinc kernel (G), the real Λ -kernel ($K_{\mathbb{R}}$), and the Dirichlet-phase Λ -kernel (H), all at $N = 2000$. The Λ -kernel exhibits strong small-spacing suppression ($P(s < 0.1) = 0.045$) lying between GOE (~ 0.1) and GUE (~ 0.01), without conforming to either universality class.

The small-spacing probability $P(s < 0.1) = 0.045$ is dramatically suppressed relative to GOE and lies between GOE and GUE. However, the spacing distribution does not conform to any standard Wigner–Dyson form: the KS distances to both GOE and GUE are comparable (~ 0.28), and the histogram exhibits a sharp peak at $s \approx 0.9$. This indicates strong *arithmetic*

rigidity—the level repulsion is real, but the long-range correlations are governed by the discrete structure of prime sums rather than by Gaussian matrix ensembles.

Dirichlet phase experiment. We further tested a complex Hermitian version $H_{ij} = K_{ij} \cdot \chi(p_i) \overline{\chi(p_j)}$ where χ is a Dirichlet character modulo 7. The spectra of $K_{\mathbb{R}}$ and H are **identical** (numerically to machine precision). The reason is structural: $\chi(p_i) \overline{\chi(p_j)}$ is a product-type phase, hence $H = D_{\chi} K D_{\chi}^{-1}$ where $D_{\chi} = \text{diag}(\chi(p_1), \dots, \chi(p_N))$. This is a unitary equivalence and leaves all eigenvalues invariant. Thus, while Dirichlet phases change the matrix representation from real symmetric to complex Hermitian, they do not alter the spectral statistics because they act by unitary conjugation rather than by genuine non-commuting perturbation.

Positive control. As a methodological check, we verified that a random anti-symmetric perturbation $H = G + i\epsilon R$ (with R a GOE matrix) instantly raises the effective rank to $O(N)$ and produces clear GUE statistics ($P(s < 0.1) = 0.000$, within statistical error consistent with GUE; KS/GUE = 0.066). This confirms that our statistical pipeline is fully functional and that the bottleneck in arithmetic perturbations is the structural nature of the arithmetic factors, not the detection methodology.

Interpretation. The Λ -kernel reveals a new regime of deterministic spectral statistics—an “arithmetic intermediate state” characterized by:

- High effective rank ($O(N)$, not $O(\log N)$);
- Strong small-spacing suppression (level repulsion beyond GOE);
- Non-Wigner–Dyson global correlations (arithmetic rigidity);
- Invariance under product-type unitary phases.

This establishes that deterministic arithmetic matrices can produce spectral statistics that deviate systematically from standard universality classes, opening a new direction in the spectral theory of arithmetic matrices. A detailed effect decomposition (discrete value-set contribution vs. non-negativity contribution) is carried out in [8], and a systematic comparison across five arithmetic kernels is presented in [9].

6 Arithmetic injection barrier

We test whether arithmetic information can be “injected” into the matrix without destroying the spectral universality. Defining

$$G_{ij}^{\mu} = \text{sinc}(\pi(t_i - t_j)) \cdot \mu(i) \mu(j),$$

where μ is the Möbius function, we find that the resulting matrix has full numerical rank but its level-spacing statistics are completely degenerate: the unfolded median spacing collapses to ~ 0.05 (versus 0.87 for GOE), and the kurtosis blows up to ~ 17 (versus 3.08 for GOE).

The algebraic root cause is that any product-type factor $\alpha(i)\alpha(j)$ introduces long-range correlations that destroy the local independence required for Wigner–Dyson universality. Combined with the Dirichlet phase result (§5.5), we have a complete picture: multiplicative arithmetic factors either freeze the spectrum (real case, e.g. Möbius) or leave it invariant (complex unimodular case, e.g. Dirichlet characters). In neither case do they steer the statistics from GOE toward GUE.

Conjecture 6.1 (Arithmetic injection barrier). *Let $\alpha : \mathbb{N} \rightarrow \mathbb{R}$ be a non-constant arithmetic sequence (e.g. Möbius, Liouville, or divisor function). Then the matrix $G_{ij}^{\alpha} = \text{sinc}(\pi(t_i - t_j)) \cdot \alpha(i)\alpha(j)$ does not exhibit GOE or GUE spectral statistics. More generally, any product-type arithmetic factor either freezes the spectrum (if the factor is real-valued and non-unimodular) or leaves it invariant (if the factor is a unimodular phase). In neither case does it steer the statistics from GOE toward GUE.*

7 Conclusion and open problems

We have introduced and analyzed a sinc-kernel Gram matrix on the primes, proving that its effective rank is $\Theta(\log N)$ by two independent arguments and that its eigenvectors are exponentially localized on frequency boxes. Numerical experiments reveal geometric spectral decay with a universal ratio $r^* \approx 1.94$ and GOE level-spacing statistics. Additionally, we have discovered that a non-convolution-type arithmetic matrix based on $\Lambda(p_i + p_j)$ breaks the effective-rank barrier with $O(N)$ positive eigenvalues and exhibits a novel “intermediate” spectral statistic, establishing that deterministic arithmetic structures can produce universality behavior beyond the standard Wigner–Dyson classes.

Several fundamental questions remain open:

1. **Exact value of the geometric decay constant.** Theorem 4.2 proves the existence of $r > 1$. Determining the exact limiting ratio $\lim_{N \rightarrow \infty} \lambda_k / \lambda_{k+1} = r^*$ analytically remains open. The numerically observed value $r^* \approx 1.94$ provides a benchmark for any future analytic determination.
2. **The universal ratio r^* .** Determine r^* analytically. The effective equation $\lambda \phi_m = V_m \phi_m + \sum_{m' \neq m} J_{mm'} \phi_{m'}$ with $V_m \sim e^m / m$ and $J_{mm'}$ from oscillatory integral estimates is the starting point.
3. **The “arithmetic intermediate state”.** Rigorously characterize the spectral statistics of the $\Lambda(p_i + p_j)$ kernel. Prove that its effective rank is $\Theta(N)$ (which first requires a rigorous proof that the Λ -kernel matrix is of full or near-full rank; this is not obvious given the sparsity of $\Lambda(p_i + p_j)$) and that its small-spacing suppression satisfies $P(s < 0.1) \rightarrow c$ for some universal constant $c \in (0, 0.1)$. Determine whether this statistic belongs to a new universality class governed by additive number theory.
4. **GOE $\rightarrow$ GUE transition.** Identify a natural arithmetic structure that genuinely breaks time-reversal symmetry without being a product-type phase. The barrier results (§6) show that neither real multiplicative factors nor complex unitary phases suffice. A genuinely non-factorizable complex Hermitian structure—perhaps based on non-abelian class field theory or higher-rank automorphic forms—may be required.
5. **Connection to ζ -zeros.** Explore whether the Λ -kernel or its generalizations bear any formal relation to the hypothetical Hilbert–Pólya operator for the Riemann zeta function. The fact that $\Lambda(p_i + p_j)$ couples primes through their *sum* (rather than ratio) suggests a connection to the additive structure of the zeta function’s Dirichlet series representation.

A The Slepian–Landau–Pollak theorem

We state the precise form of the theorem used in Section 4.

Theorem A.1 ([6, Eq. (56)–(58)]). *Let $\{x_1, \dots, x_N\} \subset [-T/2, T/2]$ and define the $N \times N$ matrix $K_{ij} = \frac{\sin(2\pi W(x_i - x_j))}{\pi(x_i - x_j)}$. For any $\delta \in (0, 1)$,*

$$\#\{k : \lambda_k(K) > \delta\} \leq 2WT + \frac{1}{\pi^2} \log\left(\frac{1 - \delta}{\delta}\right) \log(4\pi WT) + o(1).$$

In particular, for fixed δ and $WT \rightarrow \infty$, the right-hand side is $2WT + O(\log(WT) \log(1/\delta))$.

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